

MDM Mini Project Report

StudXchange – Student Exchange and Marketplace System

Team Members

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Problem Statement

Students often have unused books, notes, electronics, and other items but lack a proper platform to exchange them within campus. This leads to wastage of resources and limited communication among students.

Description

StudXchange is a web-based platform that allows students to buy, sell, and rent items within their campus community. It includes real-time chat functionality for direct communication between users, a Committee Hub for college clubs and events, and a Rent & Borrow section for short-term item usage.

Technology Used

Layer	Technologies
Client Side	HTML, CSS, JavaScript
Server Side	Node.js, Express.js, Socket.io
Database	MongoDB

Layout Design / Snapshots

Home Page

The Home Page is the main landing page of StudXchange. It showcases the platform's core value proposition and gives quick access to Marketplace, Rent & Borrow, and Committee Hub. It explains how the platform works in three steps and highlights VCET-only access, always-free usage, and full device responsiveness.

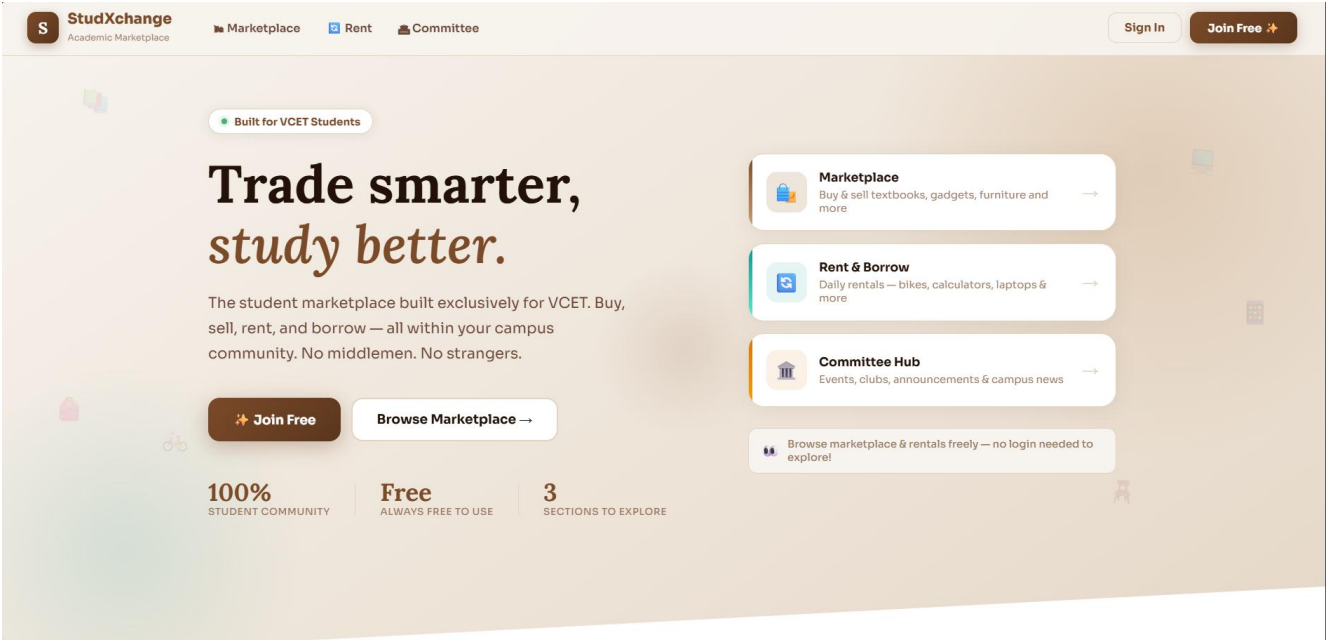


Fig 1.1 – Hero Section (Trade smarter, study better)

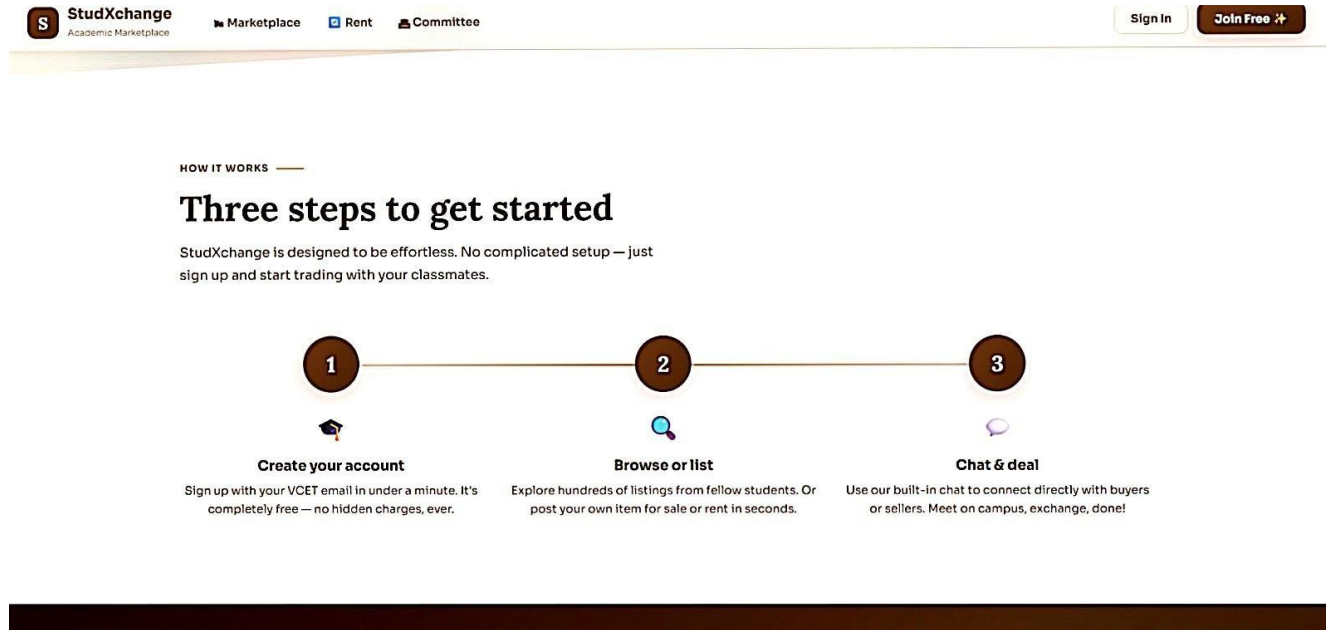


Fig 1.2 – How It Works (Three steps to get started)

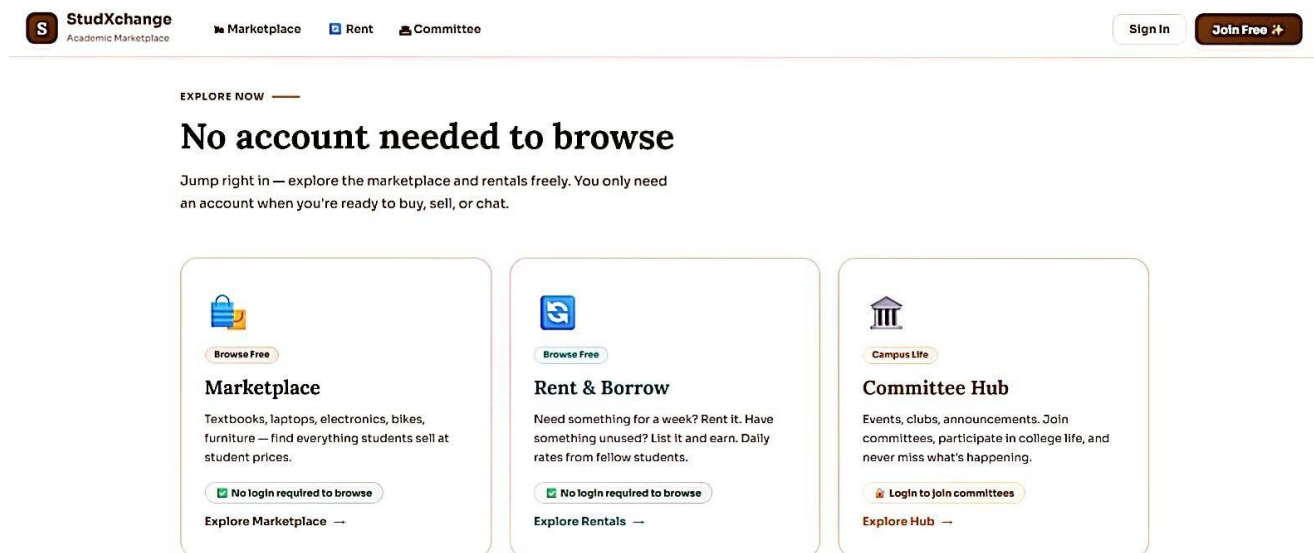


Fig 1.3 – Explore Sections (No account needed to browse)

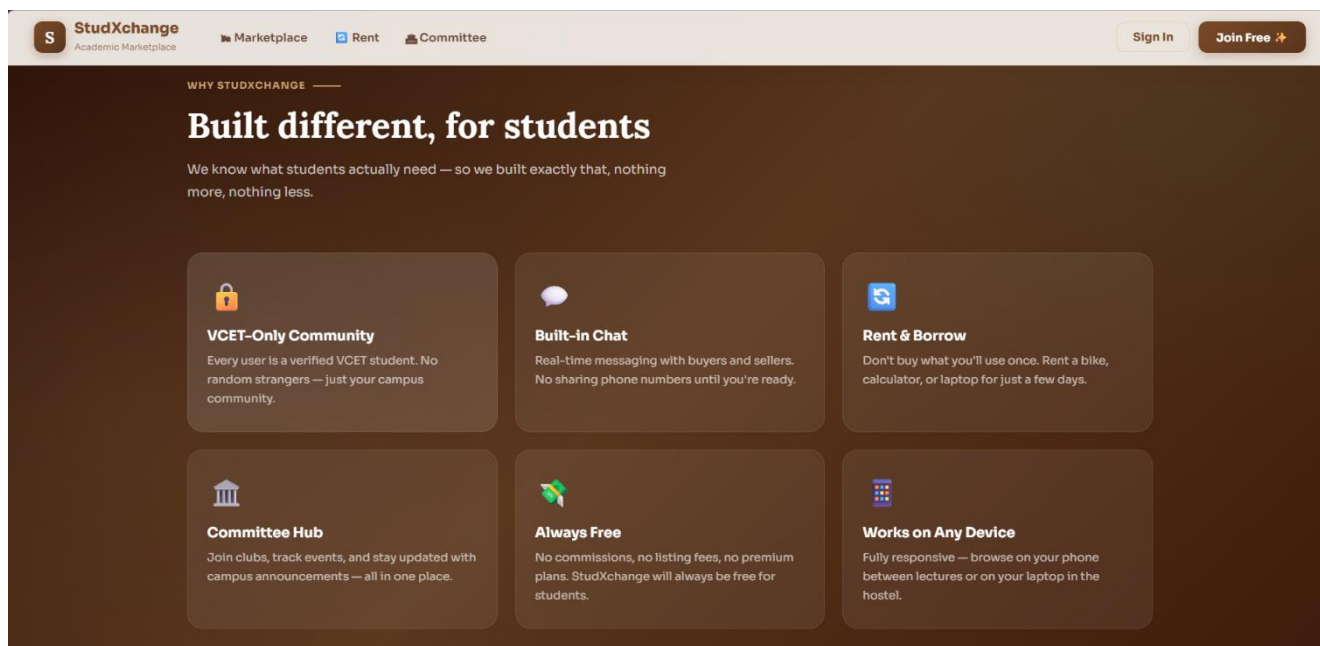


Fig 1.4 – Why StudXchange (Built different, for students)

Login / Register Page

Allows VCET students to sign in using their college email. A split-screen layout reinforces branding on the left and presents the sign-in form on the right. New users can create an account directly from this page.


StudXchange
 Academic Marketplace

Trade smarter, study better.

The student marketplace built for VCET. Buy, sell, rent — all in one place.

- Only VCET students
- Built-in chat
- Rent & borrow
- Verified community

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Welcome back 🙌

Sign in to your StudXchange account

EMAIL

PASSWORD

Sign In

Don't have an account? [Create one](#) —


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Join StudXchange 🎓

Create your account with your VCET email

FULL NAME

VCET EMAIL

PASSWORD

Create Account

Already have an account? [Sign in](#) —

Fig 2 – Login / Register Page

Marketplace Page

Central hub for buying and selling. Students browse listings by category (Laptops, Books, Electronics, Notes, Bikes, Furniture), sort by newest first, and post new items directly from this page.


StudXchange
 Academic Marketplace

+ Sell

S saniya



Marketplace

Buy & sell with your fellow students

→



Rent & Borrow

Rent items, save money together

→



Committee Hub

Events, clubs & announcements

→

All

Laptops

Books

Electronics

Notes

Bikes

Furniture

Newest Listings

1 item found

Books



Newest First

Fig 3 – Marketplace Page

Product Details Page

Displays full item information including title, price, category, seller, condition, and description. Buyers can initiate contact with the seller via built-in chat.

Sell / Post Item Page

Lets students list items for sale by filling in title, price, category, condition, description, and an optional photo.

Designed for quick listing with minimal friction.

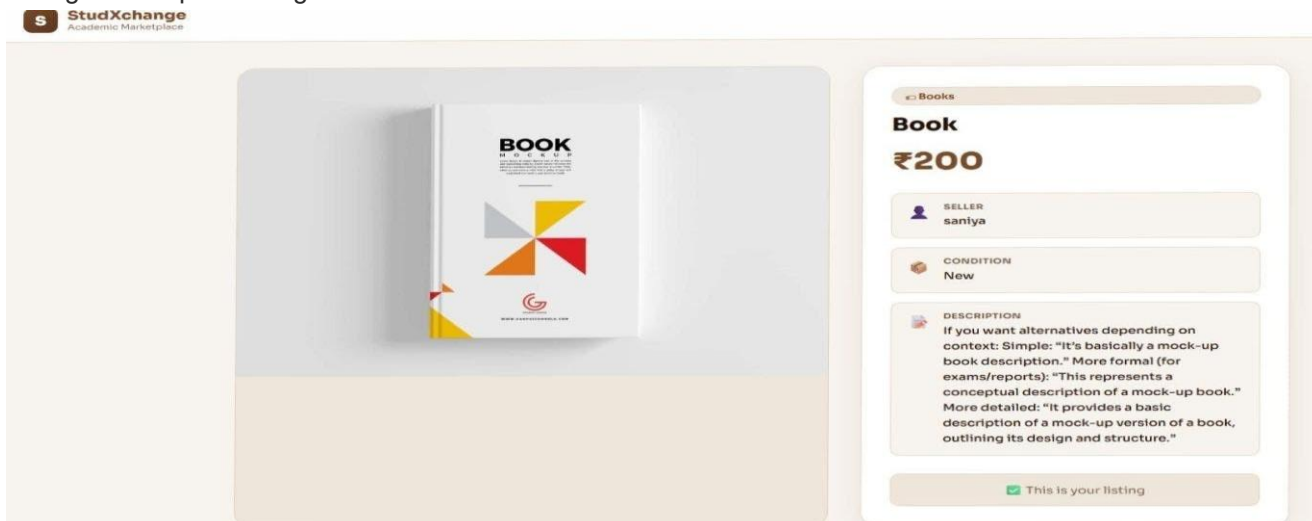


Fig 4 – Product Details Page (Buy)

The screenshot shows the 'Post Your Item' form. At the top, it says 'Post Your Item' and 'Fill in the details to list your item for sale'. Below this is a dashed box for 'UPLOAD PHOTO' with a camera icon and the text 'Click to upload image' and 'JPG, PNG up to 5MB'. Below the photo box are two rows of input fields. The first row has 'TITLE *' with the example 'e.g. Engineering Textbook' and 'PRICE (₹) *' with the example 'e.g. 350'. The second row has 'CATEGORY' with a dropdown menu showing 'Books' and 'CONDITION' with a dropdown menu showing 'New'. Below these is a 'DESCRIPTION' section with a text area and the placeholder 'Describe your item — age, usage, any defects...'. At the bottom is a 'CONTACT NUMBER' field with the example '+91 98765 43210'. At the very bottom is a large orange button with a rocket icon and the text 'Post Item'.

Fig 5 – Sell / Post Item Page

Rent & Borrow Page

Students rent out items they own or borrow what they need at student-friendly rates. Categories include Books, Laptops, Electronics, Bikes, and Furniture.

Chat Interface

Real-time messaging using Socket.io. Each conversation is tied to a specific listing. Users connect directly without sharing personal phone numbers.




No rentals yet
Be the first to list something for rent!

Fig 6 – Rent & Borrow Page

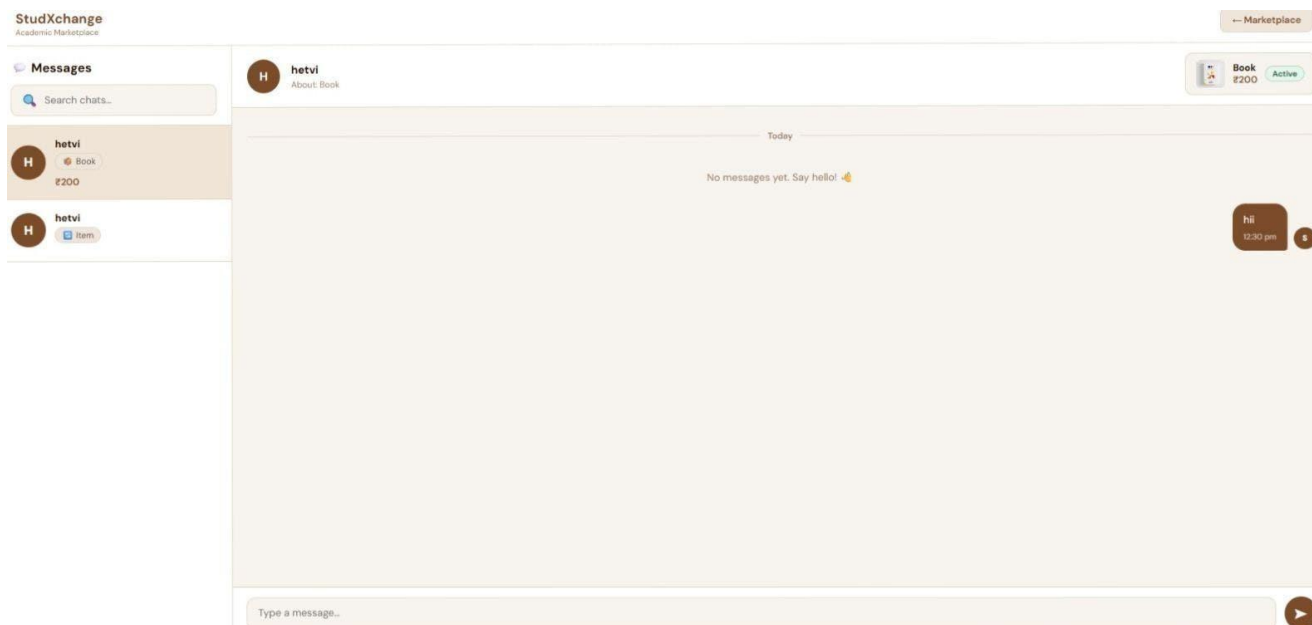


Fig 7 – Chat Interface (Real-Time Messaging)

My Profile Page

Displays a student's personal info, active listings, incoming requests, and wishlist. The Verified badge confirms VCET student identity.

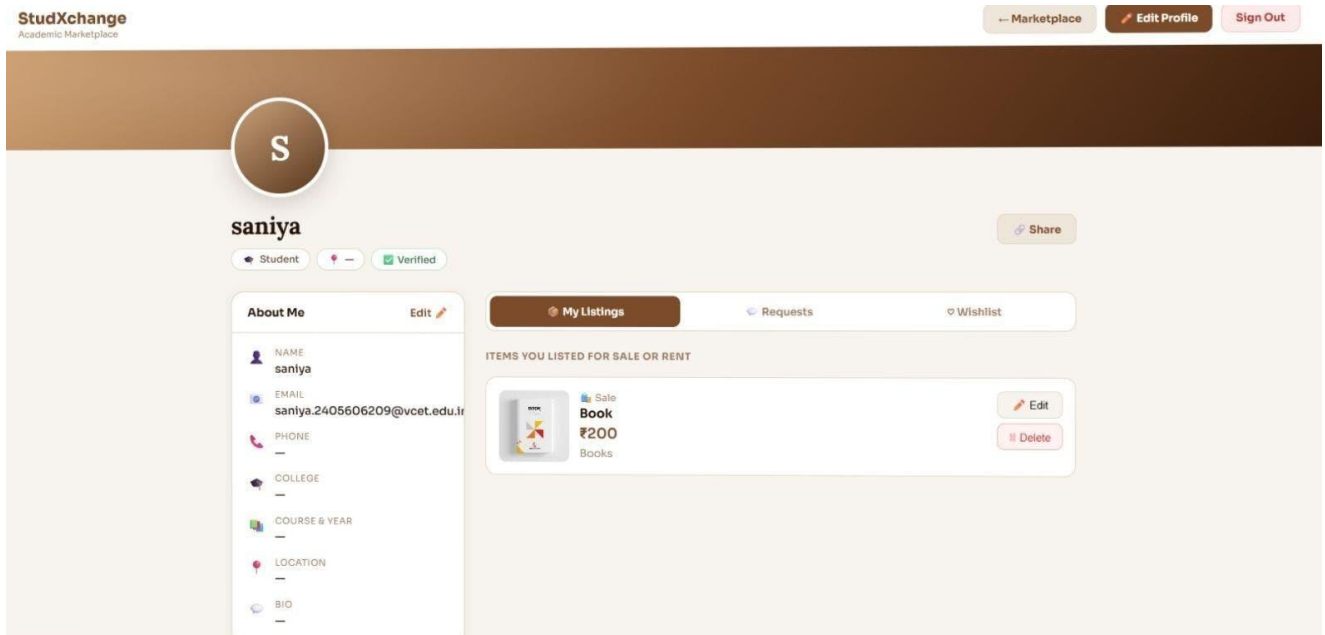


Fig 8 – My Profile Page

Committee Hub

Dedicated section for campus life — discover and join committees like Coding Club, Cultural Committee, NSS, Sports Club, Photography Club, and Entrepreneurship Cell. Also tracks events, announcements, and activity feeds.

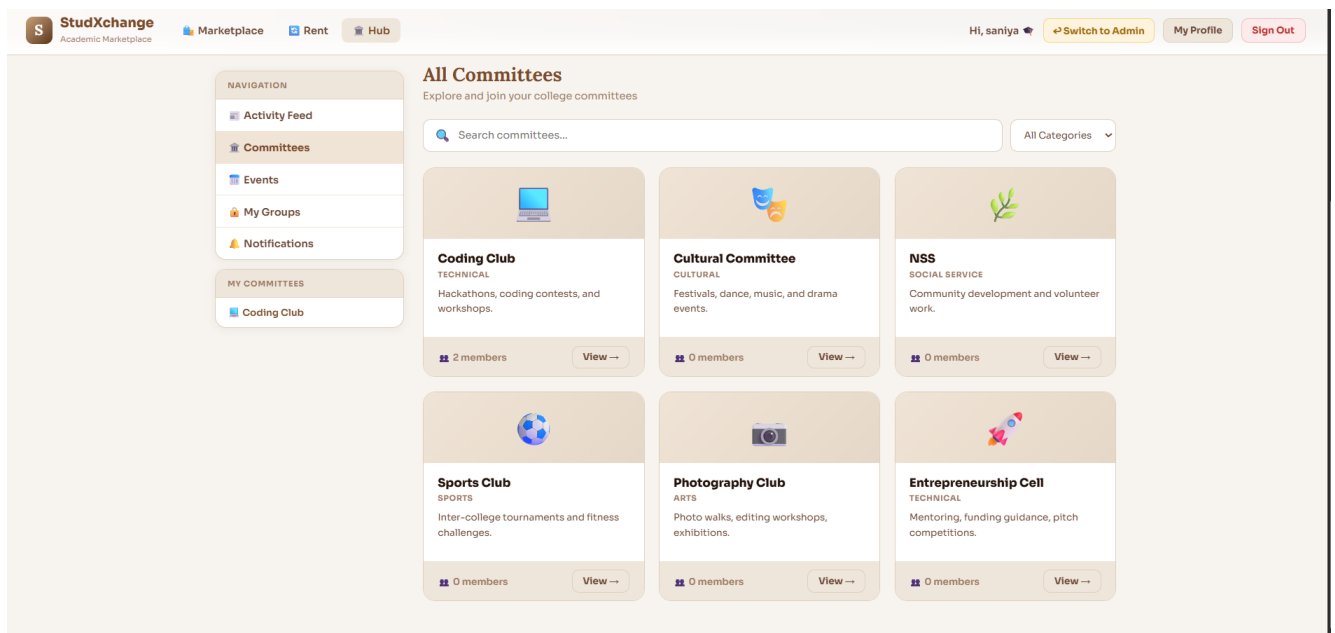


Fig 9 – Committee Hub (All Committees)

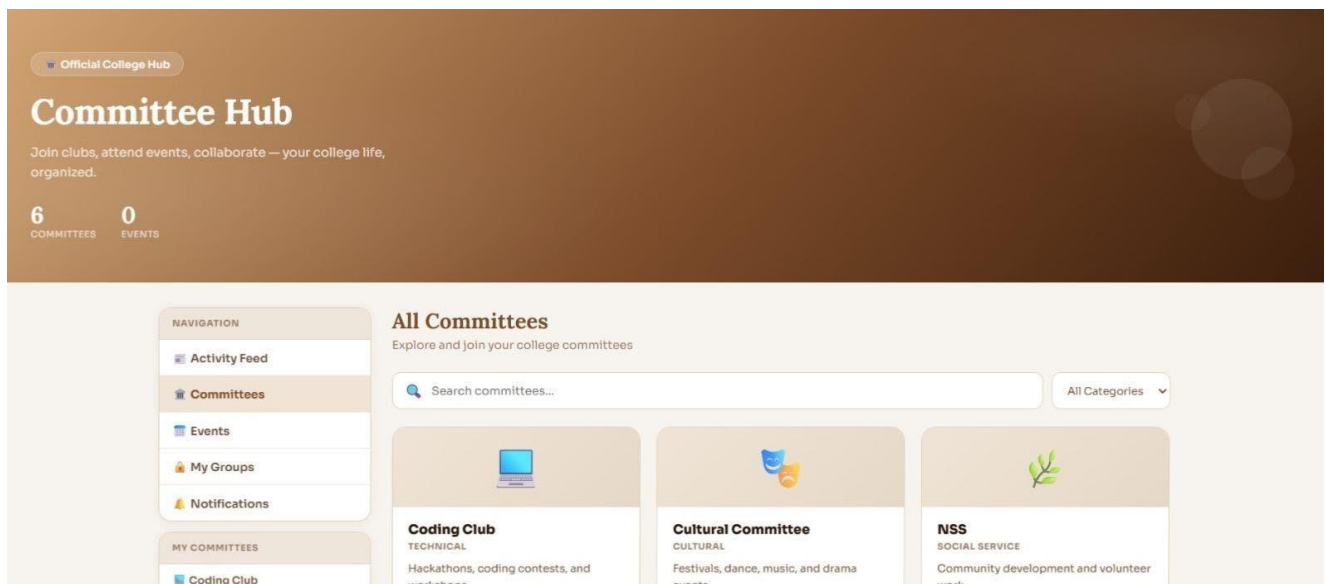


Fig 10 – Committee Hub Landing Page with Stats & Navigation

Data Base Design

The database is designed using MongoDB and follows a NoSQL document-based model. Instead of tables, data is stored in collections, and each record is a document in JSON format. Each document contains a unique `_id` field that acts as the primary identifier. Relationships between data are managed using embedded documents or references rather than foreign keys. Collections include User, Item, Message, Committee, Wishlist, and Rent Request. The ER diagram represents the logical structure of these entities.

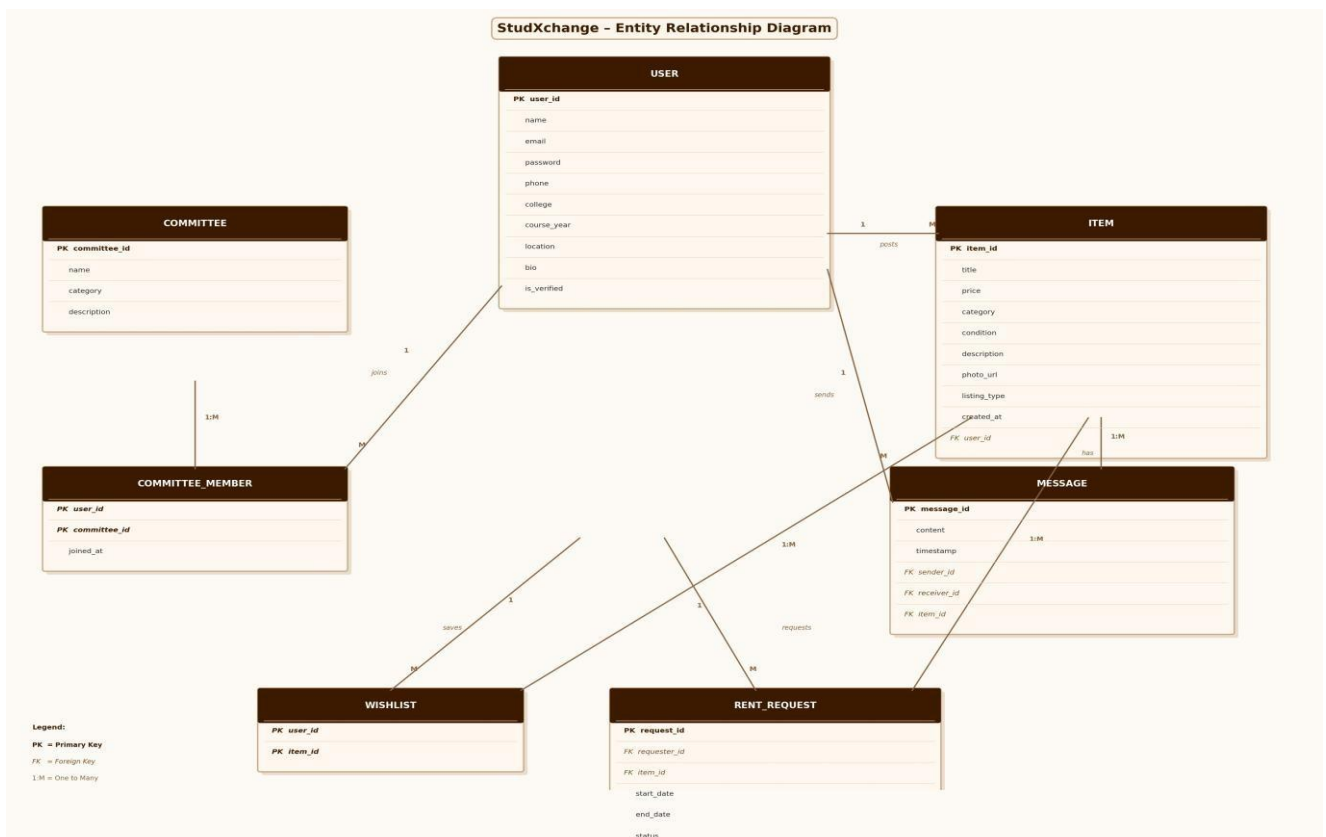


Fig 11 – Entity Relationship Diagram (StudXchange Database)

Conclusion

StudXchange provides an efficient, student-first platform for exchanging resources within a campus community. By combining a buy/sell Marketplace, a Rent & Borrow system, real-time chat, user profiles, and a Committee Hub into a single web application, the project demonstrates full-stack development skills using modern technologies. The platform reduces resource wastage, eliminates the need for third-party marketplaces, and fosters stronger communication and collaboration within the student community.